



PAN3740 系列

产品说明书

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上海磐启微电子有限公司

Sub-1G & 2.4G 多模无线 SOC 芯片

Overview

The PAN3740 series is a dual-band transceiver SOC chip that integrates Sub-1G and 2.4G.

The Sub-1G band is a low-power, long-distance wireless transceiver chip that uses ChirpIoT™ modulation and demodulation technology. It supports half-duplex wireless communication and operates in the frequency bands of 137~187MHz, 205~281MHz, 408~550MHz and 816~1110MHz. It has the characteristics of high anti-interference, high sensitivity, low power consumption and ultra-long transmission distance. It has a maximum sensitivity of -143dBm and a maximum output power of 20dBm, generating an industry-leading link budget, making it the best choice for long-distance transmission and applications with extremely high reliability requirements.

Compared with conventional modulation technologies, PAN3740 also has significant advantages in blocking and adjacent channel selection, which can further improve communication reliability. At the same time, it also provides greater flexibility. Users can adjust the spread spectrum modulation bandwidth, spread spectrum factor and error correction rate by themselves, effectively improving the compromise between distance, anti-interference ability and power consumption of chips using conventional modulation technology.

The 2.4G band supports BLE5.3 and 2.4GHz dual-mode working modes, and works in the world-wide ISM band of 2.400 ~ 2.483GHz.

At the same time, the PAN3740 series has built-in 512KB Flash program memory and 48KB SRAM memory. In addition, it is also equipped with a wealth of peripherals, including up to 19 GPIOs, 8 PWMs, 1 32-bit timer, 2 24-bit timers, 1 I2C, 2 UARTs, 2 SPIs, 7 external channels of ADC, WDT, WWD, 32K RC automatic calibration, etc.

Key Features

Sub-1G band

Working frequency band:

137~187MHz 205~281MHz 408~550MHz
816~1110MHz

Modulation : ChirpIoT™

Transmit output power: -30dBm~20dBm

Maximum link budget: 163dB

Sensitivity as low as: -143dBm@62.5kHz

Operating current

- Sleep current: 260nA

Receive current: 4.1mA@DCDC mode

- Transmit current:

96mA@20dBm

◦ 13mA@-2dBm

Supported bandwidth

- 62.5kHz, 125kHz, 250kHz, 500kHz

Support SF (5~12), automatic SF identification

Support coding rate: 4/5, 4/6, 4/7, 4/8

Support CAD function

Support low rate mode: 0.08~59.9kbps

4-wire SPI / 3-wire SPI / I2C configuration interface,
4 GPIO

2.4G band

RF

- Operating frequency band: 2.400 ~ 2.483GHz

- Supported modes

◦ Various modes of BLE5.3: 1Mbps, 2Mbps,
500kbps, 125kbps

◦ 2.4G private protocol, supports 1Mbps, 2Mbps,
500kbps, 250kbps, 125kbps, supports hardware ACK

Transmit power: up to 9dBm

Receiving sensitivity:

◦ -99dBm@125kbps

◦ -99dBm@500kbps

◦ -96dBm@1Mbps

◦ -93dBm@2Mbps

- RSSI

◦ Resolution: 0.25dB

◦ Accuracy: ±2dB

◦ Range: -90dBm ~ -15dBm

- Support single antenna

Safety regulations

MCU

- 32 位 MCU, 最高主频 48MHz

内存

- Flash: 内置 512KB, 支持深度睡眠模式

- SRAM: 48KB

- eFuse: 128B

- I-cache: 4KB

低功耗

- 接收模式: 2.5mA@1Mbps (DCDC)

- 发射模式: 5.06mA@0dBm (DCDC)

- 待机模式: 0.28uA

- 待机模式 (SRAM 保留): 1.88uA (支持 RCL /
XTL 和 GPIO 唤醒)

- 深度睡眠模式: 3.37uA (All Logic Retention /
GPIO / XTL / RCL 可唤醒)

时钟

- 32MHz RC

- 32MHz XTAL

- 32kHz RC

- 32.768kHz XTAL

- DPLL (48MHz)

外设

- 高达 21 个 GPIO (IO 有两路供电电压)

- 8 路 PWM
- 1 个 32 位定时器，2 个 24 位定时器
- 1 路 I2C
- 2 路 UART
- 2 路 SPI
- 2 路 DMA
- 12 个通道 ADC（7 个 ext, VBG_1P2, VBG_VT, 1/4VDD, VBG_0P6, Temp2）
- 支持 WDT/WWDT
- 支持 IO/BOD/POR/LVR/System 复位
- FMC（支持 IAP，支持地址为 0x0 的引导加载程序）
- 时钟测量，时钟校准
- Flash 数据加密

- 温度传感器

- 支持温度传感器
- 检测范围：-40°C ~ 85°C

封装

- QFN48

工作条件

- 工作温度：-40°C ~ 85°C
- 工作电压：2V ~ 3.6V（DCDC）

典型应用

- 安防、消防、医疗、表计、位置、传感等

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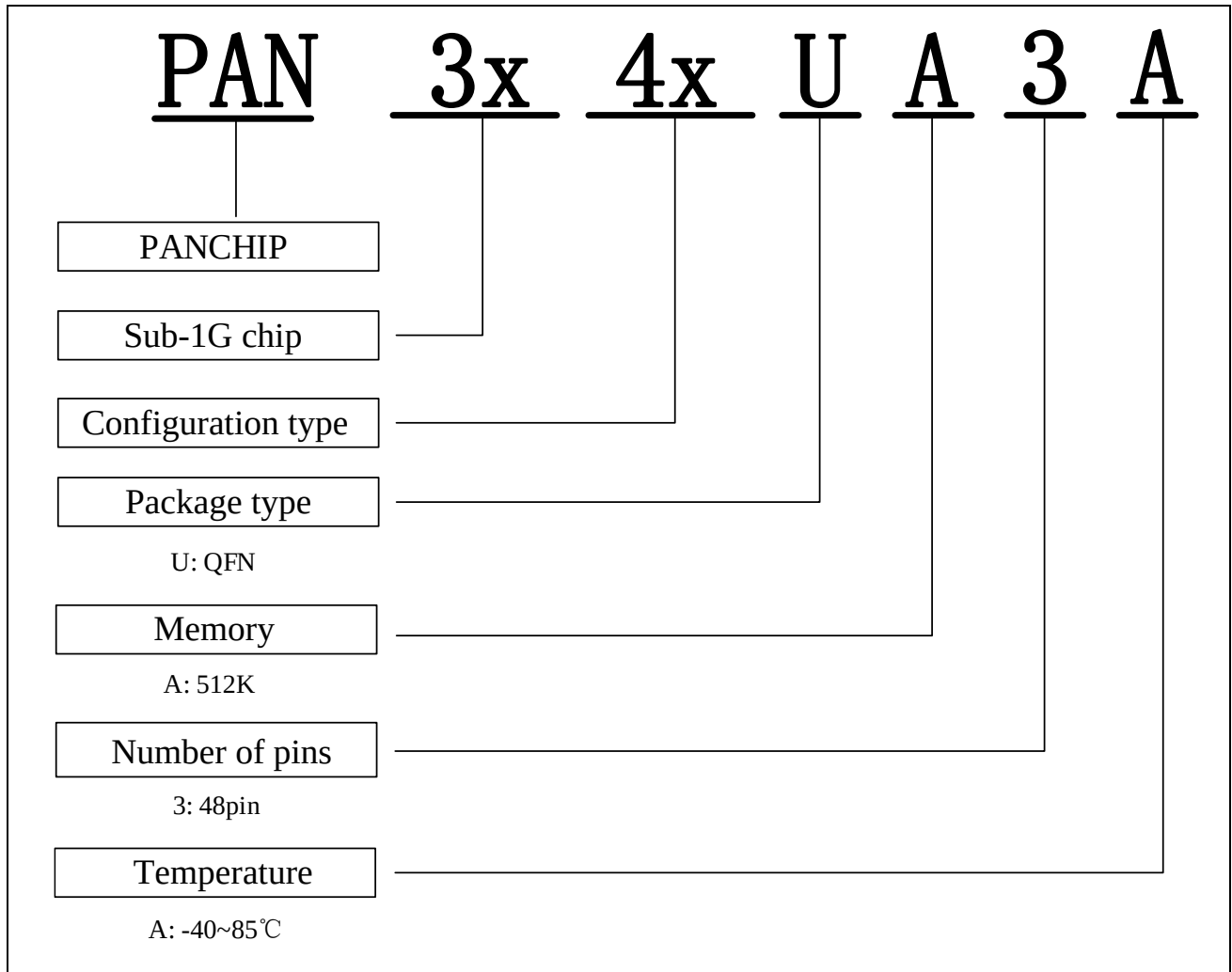
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1 命名规则



2 订购信息

产品型号	芯片类型	封装	引脚数	IO	FLASH	RAM	温度	包装
PAN3740UA3A	Sub-1G	QFN	48	21	512K	48K	-40~85°C	Tape & Reel

订购前，请咨询销售以获取最新的量产信息。

3 系统结构框图

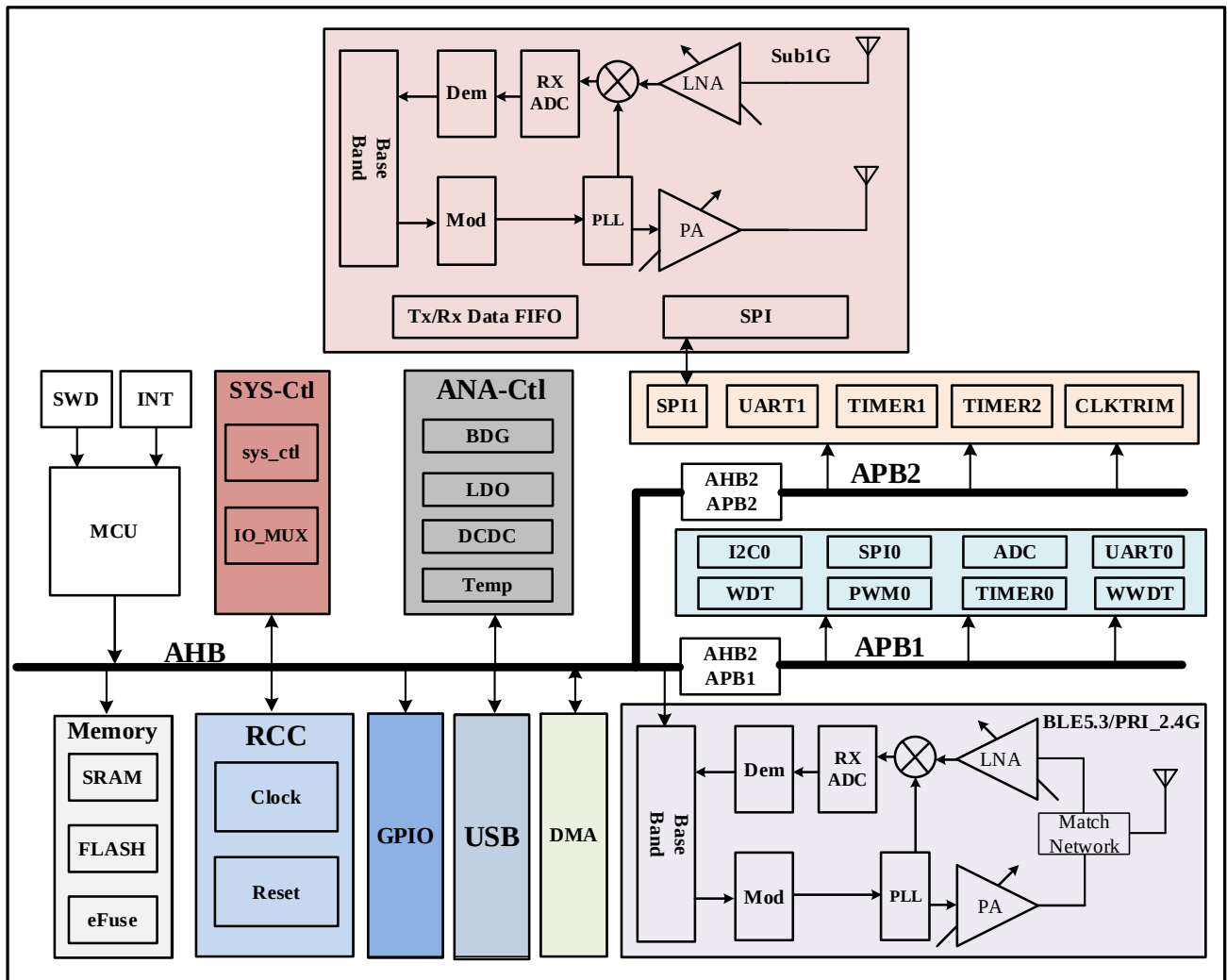


图 3-1 系统结构框图

4 引脚定义和说明

4.1 引脚图

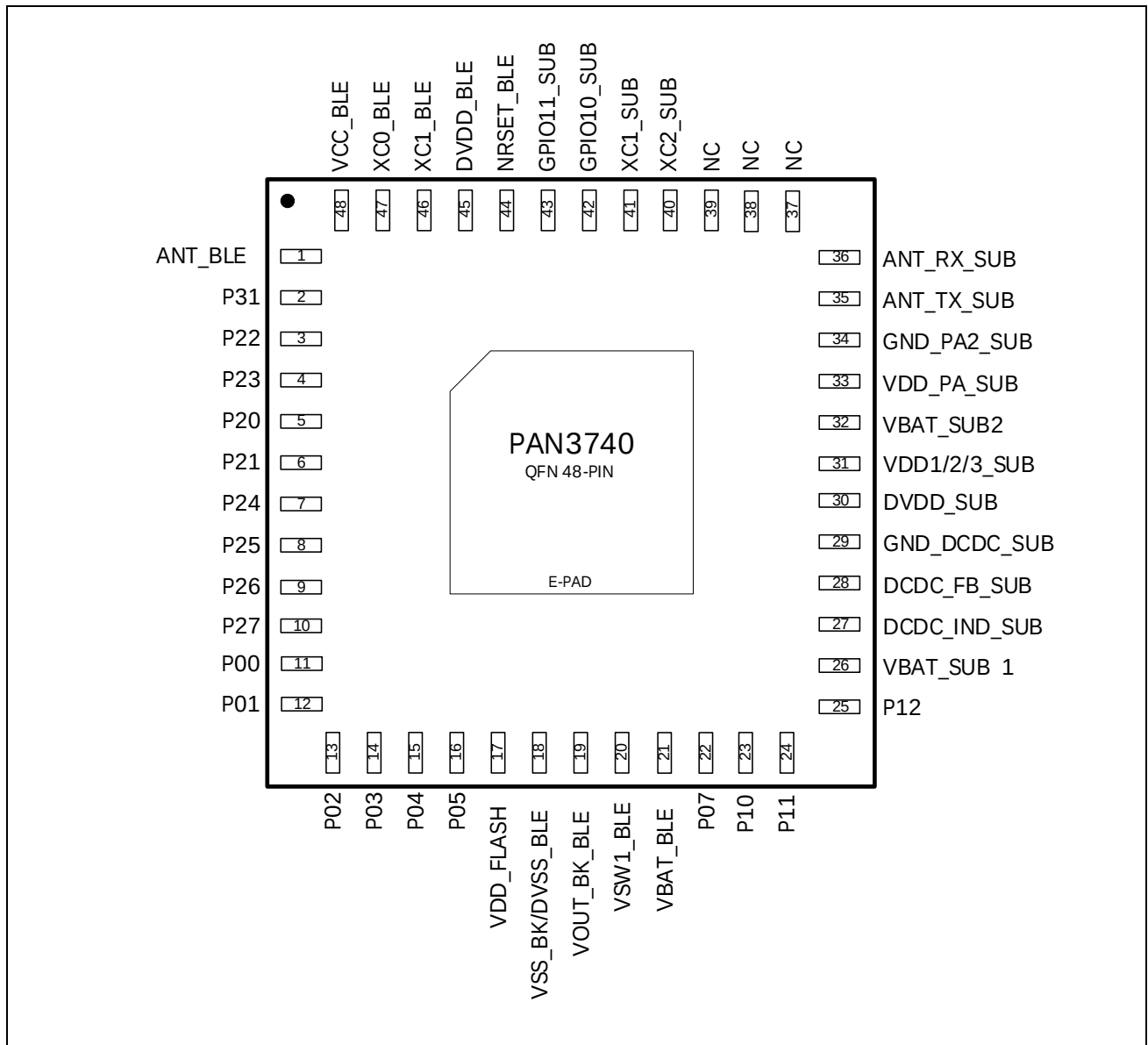


图 4-1 QFN48 引脚图

4.2 引脚说明

表 4-1 引脚说明

引脚编号	引脚名称	引脚类型	描述
1	ANT_BLE	AI/AO	射频天线引脚，使用时需要外接天线
2	P31	I/O	通用数字输入输出引脚
	UART1_TX	O	UART1 TX 引脚
	PWM_CH7	O	PWM 通道 7 输出引脚
	SPI1_MOSI	I/O	SPI1 MOSI 引脚
3	P22	I/O	通用数字输入输出引脚
	SPI1_CLK	I/O	SPI1 时钟引脚
	PWM_CH0	O	PWM 通道 0 输出引脚
	SPI0_MOSI	I/O	SPI0 MOSI 引脚
	ADC_CH2	AI	ADC 通道 2 引脚
4	P23	I/O	通用数字输入输出引脚
	SPI1_CS	I/O	SPI1 CS 引脚
	PWM_CH1	O	PWM 通道 1 输出引脚
	DPLL_DIV8	O	Dpll_div8 输出引脚
	I2C0_SCL	I/O	I2C0 SCL 引脚
	ADC_CH1	AI	ADC 通道 1 引脚
5	P20	I/O	通用数字输入输出引脚
	SPI1_MISO	I/O	SPI1 MISO 引脚
	PWM_CH6	O	PWM 通道 6 输出引脚
6	P21	I/O	通用数字输入输出引脚
	XTL0	AI	外部 32.768kHz 时钟源输出
	SPI1_MOSI	I/O	SPI1 MOSI 引脚
	PWM_CH7	O	PWM 通道 7 输出引脚
7	P24	I/O	通用数字输入输出引脚
	UART1_RX	I	UART1 RX 引脚
	SPI1_MISO	I/O	SPI1 MISO 引脚
	PWM_CH2	O	PWM 通道 2 输出引脚
	TIMER1_CNT_OUT	O	TIMER1 输出引脚
	ADC_CH6	AI	ADC 通道 6 引脚
8	P25	I/O	通用数字输入输出引脚

引脚编号	引脚名称	引脚类型	描述
	UART1_TX	O	UART1 TX 引脚
	SPI1_MOSI	I/O	SPI1 MOSI 引脚
	PWM_CH3	O	PWM 通道 3 输出引脚
	I2C0_SDA	I/O	I2C0 SDA 引脚
	ADC_CH7	AI	ADC 通道 7 引脚
9	P26	I/O	通用数字输入输出引脚
	UART1_RTS	O	UART1 RTS 引脚
	PWM_CH4	O	PWM 通道 4 输出引脚
	SPI1_CS	I/O	SPI1 CS 引脚
10	P27	I/O	通用数字输入输出引脚
	UART1_CTS	I	UART1 CTS 引脚
	PWM_CH5	O	PWM 通道 5 输出引脚
	SPI1_CLK	I/O	SPI1 时钟引脚
	XTH	O	Xth 输出引脚
11	P00	I/O	通用数字输入输出引脚
	SWD_CLK	I	SWD 时钟输入引脚
	UART1_RX	I	UART1 RX 引脚
	I2C0_SCL	I/O	I2C0 SCL 引脚
	SPI0_CLK	I/O	SPI0 时钟引脚
12	P01	I/O	通用数字输入输出引脚
	SWD_DAT	I/O	SWD 数据输入输出引脚
	UART1_TX	O	UART1 TX 引脚
	I2C0_SDA	I/O	I2C0 SDA 引脚
	SPI0_CS	I/O	SPI0 CS 引脚
13	P02	I/O	通用数字输入输出引脚
	UART0_CTS	I	UART0 CTS 引脚
	SPI1_MISO	I/O	SPI1 MISO 引脚
	PWM_CH3	O	PWM 通道 3 输出引脚
	UART1_CTS	I	UART1 CTS 引脚
14	P03	I/O	通用数字输入输出引脚
	UART0_CTS	I	UART0 CTS 引脚
	SPI0_CS	I/O	SPI0 CS 引脚
	PWM_CH2	O	PWM 通道 2 输出引脚

引脚编号	引脚名称	引脚类型	描述
	TIMER0_CNT_OUT	O	TIMER0 输出引脚
15	P04	I/O	通用数字输入输出引脚
	UART0_RTS	O	UART0 RTS 引脚
	SPI0_CLK	I/O	SPI0 时钟引脚
	PWM_CH3	O	PWM 通道 3 输出引脚
	TM0_EXT	I	TIMER0 外部输入引脚
	ADC_CH0	AI	ADC 通道 0 引脚
16	P05	I/O	通用数字输入输出引脚
	UART0_TX	O	UART0 TX 引脚
	SPI0_MISO	I/O	SPI0 MISO 引脚
	PWM_CH4	O	PWM 通道 4 输出引脚
17	VDD_FLASH	P	FLASH 电源
18	DVSS_BLE	P	数字地
	VSS_BK_BLE	P	DCDC 电源的公共接地端，独立的电源地
19	VOUT_BK_BLE	P	DCDC 电压输出
20	VSW1_BLE	P	DCDC 内部功率开关（开关频率约 650kHz），使用时需要外接电感
21	VBAT_BLE	P	芯片的电源输入引脚
22	P07	I/O	通用数字输入输出引脚
	UART1_RX	I	UART1 RX 引脚
	I2C0_SCL	I/O	I2C0 时钟引脚
	SPI0_MOSI	I/O	SPI0 MOSI 引脚
	PWM_CH0	O	PWM 通道 0 输出引脚
	ADC_CH5	AI	ADC 通道 5 引脚
23	P10	I/O	通用数字输入输出引脚
	UART1_TX	O	UART1 TX 引脚
	I2C0_SDA	I/O	I2C0 SDA 引脚
	SPI0_MISO	I/O	SPI0 MISO 引脚
	PWM_CH5	O	PWM 通道 5 输出引脚
24	P11	I/O	通用数字输入输出引脚
	UART1_RTS	O	UART1 RTS 引脚
	SPI0_MOSI	I/O	SPI0 MOSI 引脚
	PWM_CH7	O	PWM 通道 7 输出引脚

引脚编号	引脚名称	引脚类型	描述
	CLK_32K	O	Clk_32k 输出引脚
	UART0_TX	O	UART0 TX 引脚
25	P12	I/O	通用数字输入输出引脚
	UART0_RX	I	UART0 RX 引脚
	TIMER0_CNT_OUT	O	TIMER0 输出引脚
	UART1_TX	O	UART1 TX 引脚
	SPI0_MISO	I/O	SPI0 MISO 引脚
	PWM_CH4	O	PWM 通道 4 输出引脚
	ADC_CH3	AI	ADC 通道 3 引脚
26	VBAT_SUB1	P	数字 GPIO 电源 / DCDC 电源，连接总电源
27	DCDC_IND_SUB	AO	内部 DCDC 输出，连接外部串联电感
28	DCDC_FB_SUB	AI	内部 DCDC 反馈输入，DCDC 模式与 VDD1/2/3 相连
29	GND_DCDC_SUB	G	模拟地
30	DVDD_SUB	P	数字电源 LDO 输出
31	VDD1/2/3_SUB	P	模拟电源，DCDC 模式连接 VFB，非 DCDC 模式连接总电源
32	VBAT_SUB2	P	模拟电源，连接总电源
33	VDD_PA_SUB	P	PA 电源 LDO 输出
34	GND_PA2_SUB	P	PA 模拟地
35	ANT_TX_SUB	AO	发射端 PA 输出
36	ANT_RX_SUB	AI	接收端 LNA 输入
37	NC		
38	NC		
39	NC		
40	XC2_SUB	AI	晶振输入
41	XC1_SUB	AO	晶振输出
42	GPIO10_SUB	I/O	数字 IO
		O	外置 PA 控制信号
43	GPIO11_SUB	I/O	数字 IO
		O	信道状态指示信号
44	NRSET_BLE	I	复位引脚
45	DVDD_BLE	P	HLDO 输出引脚，典型值 1.2V
46	XC1_BLE	AO	外部 32MHz 时钟源输出

引脚编号	引脚名称	引脚类型	描述
47	XC0_BLE	AI	外部 32MHz 时钟源输入
48	VCC_BLE	P	芯片供电电源
49	E-PAD	P	芯片底部焊盘，公共接地端

4.3 内部连接

表 4-2 内部连接引脚说明

Pin Status	RF	MCU
I S	PAD_MOSI_3V	P16
I S	PAD_MISO_3V	P30
I S	PAD_CSN_3V	P14
I S	PAD_SCK_3V	P15
I S	PAD_IRQ_3V	P17
I S	NRSET	P13

5 电气特性

最大值和最小值

在每个表格下方的注解中说明为通过综合评估、设计模拟和/或工艺特性得到的数据，不会在生产线上进行测试；在综合评估的基础上，最小和最大数值是通过样本测试后，取其平均值再加减三倍的标准分布(平均 $\pm 3\sigma$)得到。

5.1 Sub-1G 频段

注：

- 本节所述 VDD 等参数对应引脚图中“xxx_SUB”等相关电源输入引脚。
- 本节所有参数是基于 $T_A=25^{\circ}\text{C}$ 、供电电压 3.3V 测试，除非有其他特殊说明。

5.1.1 绝对最大额定值

表 5-1 绝对最大额定值

符号	参数	条件	最小值	典型值	最大值	单位
VDD	VDD1/VDD2/VDD3/VBAT/VBAT_IO	频率：490MHz	-0.3	3.3	3.6	V
V _I	输入电压		-0.3	-	VDD	V
V _O	输出电压		VSS	-	VDD	V
T _{OP}	工作温度		-40	-	85	°C
T _{STG}	存储温度		-55	-	125	°C

注意：

- 超过一个或多个限制值可能会对芯片造成永久性损坏。
- 静电敏感设备，操作时符合保护规则。

5.1.2 直流电特性

测试条件：

- 频率：490MHz

表 5-2 电压和电流（频率 490MHz）

符号	参数	最小值	典型值	最大值	单位	测试条件
VDD	电源	1.8	3.3	3.6	V	T _A =25°C, LDO 模式
		2	3.3	3.6	V	T _A =25°C, DCDC 模式
VSS	地	-	0	-	V	-
I _{DeepSleep}	深度休眠电流	-	260	-	nA	-
I _{Sleep}	休眠电流	-	620	-	nA	-
I _{TX,20dBm}	TX 模式的工作电流	-	96	-	mA	20dBm 输出功率
I _{TX,18dBm}	TX 模式的工作电流	-	77	-	mA	18dBm 输出功率
I _{TX,-2dBm}	TX 模式的工作电流	-	13	-	mA	-2dBm 输出功率
I _{RX,LDO}	RX 模式的工作电流	-	7.8	-	mA	LDO 模式下, 最大 LNA 增益
I _{RX,DCDC}	RX 模式的工作电流	-	4.1	-	mA	DCDC 模式下, 最大 LNA 增益
V _{OH}	输出高电平电压	VDD-0.3	-	VDD	V	-
V _{OL}	输出低电平电压	VSS	-	VSS+0.3	V	-
V _{IH}	输入高电平电压	0.8*VDD	-	-	V	-
V _{IL}	输入低电平电压	-	-	0.2*VDD	V	-
SPI_rate	SPI 速率	-	-	10	Mbps	-

测试条件：

- 频率：915MHz

表 5-3 电压和电流（频率 915MHz）

符号	参数	最小值	典型值	最大值	单位	测试条件
VDD	电源	1.8	3.3	3.6	V	T _A =25°C, LDO 模式
		2	3.3	3.6	V	T _A =25°C, DCDC 模式
VSS	地	-	0	-	V	-
I _{DeepSleep}	深度休眠电流	-	260	-	nA	-
I _{Sleep}	休眠电流	-	620	-	nA	-
I _{TX,20dBm}	TX 模式的工作电流	-	100	-	mA	20dBm 输出功率

$I_{TX,18dBm}$	TX 模式的工作电流	-	81	-	mA	18dBm 输出功率
$I_{TX,-2dBm}$	TX 模式的工作电流	-	15	-	mA	-2dBm 输出功率
$I_{RX,LDO}$	RX 模式的工作电流	-	7.8	-	mA	LDO 模式下, 最大 LNA 增益
$I_{RX,DCDC}$	RX 模式的工作电流	-	4.1	-	mA	DCDC 模式下, 最大 LNA 增益
V_{OH}	输出高电平电压	VDD-0.3	-	VDD	V	-
V_{OL}	输出低电平电压	VSS	-	VSS+0.3	V	-
V_{IH}	输入高电平电压	0.8*VDD	-	-	V	-
V_{IL}	输入低电平电压	-	-	0.2*VDD	V	-
SPI_rate	SPI 速率	-	-	10	Mbps	-

5.1.3 RF 性能

测试条件:

- 频率: 490MHz
- 纠错码 = 4/8
- 误包率 ≤ 5%
- Payload 长度=10Bytes

表 5-4 通用 RF 特性 (频率 490MHz)

符号	描述	条件	最小	典型	最大	单位
通用频率						
F _{op}	工作频率	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _S	晶体串联电阻	-	-	30	50	Ω
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	±10	-	ppm
BR	比特速率	-	0.08	-	59.9	kbps
发射器						
P _{LPWAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-129 -137 -143	- - -	dBm
RF_125 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-126 -134 -140	- - -	dBm
RF_250 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-124 -132 -137	- - -	dBm
RF_500 (LDO)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -134	- - -	dBm
RF_62.5 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -133 -139	- - -	dBm
RF_125 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-122 -130 -137	- - -	dBm

RF_250 (DCDC)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-120 -128 -134	- - -	dBm
RF_500 (DCDC)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-117 -125 -130	- - -	dBm

测试条件：

- 频率：915MHz
- 纠错码 = 4/8
- 误包率 ≤ 5%
- Payload 长度 = 10Bytes

表 5-5 通用 RF 特性（频率 915MHz）

符号	描述	条件	最小	典型	最大	单位
通用频率						
F _{op}	工作频率	-	137	-	187	MHz
		-	205	-	281	MHz
		-	408	-	550	MHz
		-	816	-	1110	MHz
F _{xtal}	晶振频率	-	-	32	-	MHz
R _s	晶体串联电阻	-	-	30	50	Ω
C _{FOOT}	晶体外部电容	-	8	15	22	pF
C _{LOAD}	晶体负载电容	-	6	10	12	pF
F _{TOL}	初始频率容限	-	-	±10	-	ppm
BR	比特速率	-	0.08	-	59.9	kbps
发射器						
PLP _{WAN}	输出功率	-	-30	-	20	dBm
接收器						
RF_62.5 (LDO)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-128 -136 -141	- - -	dBm
RF_125 (LDO)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-125 -132 -138	- - -	dBm
RF_250 (LDO)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-121 -129 -135	- - -	dBm
RF_500 (LDO)	RF 灵敏度，长距离模式，最高 LNA 增益， 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7 SF = 10 SF = 12	- - -	-119 -127 -132	- - -	dBm
RF_62.5	RF 灵敏度，长距离模式，最高 LNA 增益，	SF = 7	-	-128	-	dBm

(DCDC)	使用分离的 RX/TX 通道 62.5 kHz 带宽	SF = 10	-	-136	-	
		SF = 12	-	-140	-	
RF_125 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 125 kHz 带宽	SF = 7	-	-125	-	dBm
		SF = 10	-	-132	-	
		SF = 12	-	-137	-	
RF_250 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 250 kHz 带宽	SF = 7	-	-120	-	dBm
		SF = 10	-	-129	-	
		SF = 12	-	-135	-	
RF_500 (DCDC)	RF 灵敏度, 长距离模式, 最高 LNA 增益, 使用分离的 RX/TX 通道 500 kHz 带宽	SF = 7	-	-118	-	dBm
		SF = 10	-	-126	-	
		SF = 12	-	-130	-	

5.2 2.4G 频段

注：

- 本节所述 VDD 等参数对应引脚图中“xxx_BLE”等相关电源输入引脚。

5.2.1 绝对最大额定值

表 5-6 绝对最大额定值

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VDD - VSS	Supply voltages	T _A =25°C	-0.3	-	3.6	V
VIN	I/O pin voltage	T _A =25°C	VSS-0.3	-	VDD + 0.3	V
PVDD	Extreme power consumption	VDD=3.3V, T _A =25°C DCDC power supply	-	-	TBD	mW

5.2.2 常规工作条件

表 5-7 常规工作条件

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VDD*	Operating voltage	T _A =25°C	1.8	-	3.6	V
VIPIO2*	Operating voltage	T _A =25°C	1.8	-	3.6	V
T _{ST}	Storage temperature	-	-65	-	150	°C
T _A	Ambient temperature	-	-40	-	125	°C
T _J	Junction temperature	QFN32 4x4x0.75-P0.4	-40	-	125	°C
R _{θJA}	Thermal resistance	QFN32 4x4x0.75-P0.4	-	TBD	-	°C/W

注: DCDC-OFF

5.2.3 RF 特性

表 5-8 RF 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f _{OP}	Operating frequency		2400	-	2483	MHz
PLLres	PLL programming resolution		244	1M	-	Hz
DR	Data rate		0.125	-	2	Mbps
Δf _{BLE,2M}	Frequency deviation @ BLE 2Mbps		450	500	550	kHz
Δf _{BLE,1M}	Frequency deviation @ BLE 1Mbps		225	250	275	kHz
Δf _{297,2M}	Frequency deviation @ 297mode 2Mbps		450	500	550	kHz

$\Delta f_{297,1M}$	Frequency deviation @ 297mode 1Mbps		225	250	275	kHz
$\Delta f_{N,2M}$	Frequency deviation @ N-mode 2Mbps		-	320	-	kHz
$\Delta f_{N,1M}$	Frequency deviation @ N-mode 1Mbps		-	170	-	kHz
$\Delta f_{BLE,2M}$	Channel spacing @ BLE 2Mbps		-	2	-	MHz
$\Delta f_{BLE,1M}$	Channel spacing @ BLE 1Mbps		-	2	-	MHz
$\Delta f_{297,2M}$	Channel spacing @ 297mode 2Mbps		-	2	-	MHz
$\Delta f_{297,1M}$	Channel spacing @ 297mode 1Mbps		-	1	-	MHz
$\Delta f_{N,2M}$	Channel spacing @ N-mode 2Mbps		-	2	-	MHz
$\Delta f_{N,1M}$	Channel spacing @ N-mode 1Mbps		-	1	-	MHz

表 5-9 TX 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
P_{RFTX}	Output power		-	-	9	dBm
P_{RFC}	RF power control range		-	40	-	dB
P_{RFCR}	RF power accuracy		-	-	± 3	dB
$P_{RF1M,1}$	1st Adjacent Channel Transmit Power @1Mbps		-	-40	-	dBc
$P_{RF1M,2}$	2nd Adjacent Channel Transmit Power @1Mbps		-	-56	-	dBc
$P_{RF1M,\geq 3}$	3rd Adjacent Channel Transmit Power @1Mbps		-	-60	-	dBc
$P_{RF2M,2}$	1st Adjacent Channel Transmit Power @2Mbps		-	-37	-	dBc
$P_{RF2M,4}$	2nd Adjacent Channel Transmit Power @2Mbps		-	-58	-	dBc
$P_{RF2M,\geq 6M}$	3rd Adjacent Channel Transmit Power @2Mbps		-	-60	-	dBc
P_{BW1M}	20dB bandwidth @1Mbps		-	1.3	-	MHz
P_{BW2M}	20dB bandwidth @2Mbps		-	2.3	-	MHz
$P_{SP,1}$	Spurious @ $\leq 1GHz$		-	-	-73	dBm
$P_{SP,2}$	Spurious @ $\geq 1GHz$		-	-	-49	dBm

表 5-10 RX 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
$P_{RX,MIX}$	Receive maximum input power		-	0	-	dBm
$P_{SENS,1M,BLE}$	Sensitivity, 1Mbps BLE		-	-96	-	dBm
$P_{SENS,2M,BLE}$	Sensitivity, 2Mbps BLE		-	-93	-	dBm
$P_{SENS,125K,BLE}$	Sensitivity, 125kbps BLE		-	-99	-	dBm
$P_{SENS,500K,BLE}$	Sensitivity, 500kbps BLE		-	-99	-	dBm

P _{SENS,250K,B}	Sensitivity, 250kbps B mode	Sensitivity, 1Mbps ideal transmitter, ≤ 37 bytes, BER = 0.1% is presented.	-	-99	-	dBm
P _{SENS,1M,297}	Sensitivity, 1Mbps 297 mode		-	-93	-	dBm
P _{SENS,2M,297}	Sensitivity, 2Mbps 297 mode		-	-91	-	dBm
P _{SENS,250K,297}	Sensitivity, 250kbps 297 mode		-	-99	-	dBm
P _{SENS,1M,N}	Sensitivity, 1Mbps N-mode		-	-93	-	dBm
P _{SENS,2M,N}	Sensitivity, 2Mbps N-mode		-	-91	-	dBm
P _{SENS,250K,N}	Sensitivity, 250kbps N-mode		-	-99	-	dBm
C/I _{CO,1M,BLE}	Co-Channel interference@1Mbps		-	6	-	dB
C/I _{1M,1M,BLE}	Adjacent (1 MHz) interference@1Mbps		-	-3	-	dB
C/I _{2M,1M,BLE}	Adjacent (2 MHz) interference @1Mbps		-	-40	-	dB
C/I _{$\geq 3M,1M,BLE$}	Adjacent (≥ 3 MHz) interference @1Mbps		-	-45	-	dB
C/I _{Image,1M,BLE}	Image frequency interference @1Mbps		-	-23	-	dB
C/I _{Image$\pm 1M,1M,BLE$}	Adjacent (± 1 MHz) interference to in-band image frequency @1Mbps		-	-38	-	dB
C/I _{$\geq 6M,1M,BLE$}	Adjacent (≥ 6 MHz) interference @1Mbps		-	-45	-	dB
C/I _{CO,2M,BLE}	Co-Channel interference @2Mbps		-	7	-	dB
C/I _{2M,2M,BLE}	Adjacent (2 MHz) interference @2Mbps		-	-3	-	dB
C/I _{4M,2M,BLE}	Adjacent (4 MHz) interference @2Mbps		-	-37	-	dB
C/I _{$\geq 6M,2M,BLE$}	Adjacent (≥ 6 MHz) interference @2Mbps		-	-35	-	dB
C/I _{Image,2M,BLE}	Image frequency interference @2Mbps		-	-23	-	dB
C/I _{Image$\pm 2M,2M,BLE$}	Adjacent (± 2 MHz) interference to in-band image frequency		-	-36	-	dB
C/I _{$\geq 12M,2M,BLE$}	Adjacent (≥ 12 MHz) interference @2Mbps		-	-35	-	dB
C/I _{CO,125K,BLE}	Co-Channel interference @125kbps		-	-6	-	dB
C/I _{1M,125K,BLE}	Adjacent (1 MHz) interference @125kbps		-	-13	-	dB
C/I _{2M,125K,BLE}	Adjacent (2 MHz) interference @125kbps		-	-37	-	dB
C/I _{$\geq 3M,125K,BLE$}	Adjacent (≥ 3 MHz) interference @125kbps		-	-49	-	dB
C/I _{Image,125K,BLE}	Image frequency interference @125kbps		-	-31	-	dB
C/I _{Image$\pm 1M,125K,BLE$}	Adjacent (± 1 MHz) interference to in-band image frequency @125kbps		-	-48	-	dB
C/I _{CO,500K,BLE}	Co-Channel interference @500kbps		-	1	-	dB
C/I _{1M,500K,BLE}	Adjacent (1 MHz) interference @500kbps		-	-6	-	dB
C/I _{2M,500K,BLE}	Adjacent (2 MHz) interference @500kbps		-	-31	-	dB
C/I _{$\geq 3M,500K,BLE$}	Adjacent (≥ 3 MHz) interference @500kbps		-	-43	-	dB
C/I _{Image,500K,BLE}	Image frequency interference @500kbps		-	-25	-	dB

C/I _{Image±1M,500K,BLE}	Adjacent (±1MHz) interference to in-band image frequency @500kbps		-	-41	-	dB
C/I _{CO,1M,297}	Co-Channel interference @1Mbps		-	TBD	-	dB
C/I _{1M,1M,297}	Adjacent (1 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{2M,1M,297}	Adjacent (2 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{≥3M,1M,297}	Adjacent (≥3 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{Image,1M,297}	Image frequency interference @1Mbps		-	TBD	-	dB
C/I _{Image±1M,1M,297}	Adjacent (±1MHz) interference to in-band image frequency @1Mbps		-	TBD	-	dB
C/I _{≥6M,1M,297}	Adjacent (≥6 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{CO,2M,297}	Co-Channel interference @2Mbps		-	TBD	-	dB
C/I _{2M,2M,297}	Adjacent (2 MHz) interference @2Mbps		-	TBD	-	dB
C/I _{4M,2M,297}	Adjacent (4 MHz) interference @2Mbps		-	TBD	-	dB
C/I _{≥6M,2M,297}	Adjacent (≥6 MHz) interference @2Mbps		-	TBD	-	dB
C/I _{Image,2M,297}	Image frequency interference @2Mbps		-	TBD	-	dB
C/I _{Image±2M,2M,297}	Adjacent (±2MHz) interference to in-band image frequency @2Mbps		-	TBD	-	dB
C/I _{≥12M,2M,297}	Adjacent (≥12 MHz) interference @2Mbps		-	TBD	-	dB
C/I _{CO,250K,297}	Co-Channel interference @250kbps		-	TBD	-	dB
C/I _{1M,250K,297}	Adjacent (1 MHz) interference @250kbps		-	TBD	-	dB
C/I _{2M,250K,297}	Adjacent (2 MHz) interference @250kbps		-	TBD	-	dB
C/I _{≥3M,250K,297}	Adjacent (≥3MHz) interference @250kbps		-	TBD	-	dB
C/I _{Image,250K,297}	Image frequency interference @250kbps		-	TBD	-	dB
C/I _{Image±1M,250K,297}	Adjacent (±1MHz) interference to in-band image frequency @250kbps		-	TBD	-	dB
C/I _{CO,1M,N}	Co-Channel interference@1Mbps		-	TBD	-	dB
C/I _{1M,1M,N}	Adjacent (1 MHz) interference@1Mbps		-	TBD	-	dB
C/I _{2M,1M,N}	Adjacent (2 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{≥3M,1M,N}	Adjacent (≥3 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{Image,1M,N}	Image frequency interference@1Mbps		-	TBD	-	dB
C/I _{Image±1M,1M,N}	Adjacent (±1MHz) interference to in-band image frequency @1Mbps		-	TBD	-	dB
C/I _{≥6M,1M,N}	Adjacent (≥6 MHz) interference @1Mbps		-	TBD	-	dB
C/I _{CO,2M,N}	Co-Channel interference @2Mbps		-	TBD	-	dB
C/I _{2M,2M,N}	Adjacent (2 MHz) interference @2Mbps		-	TBD	-	dB
C/I _{4M,2M,N}	Adjacent (4 MHz) interference @2Mbps		-	TBD	-	dB

$C/I_{\geq 6M,2M,N}$	Adjacent (≥ 6 MHz) interference @2Mbps		-	TBD	-	dB
$C/I_{Image,2M,N}$	Image frequency interference @2Mbps		-	TBD	-	dB
$C/I_{Image \pm 2M,2M,N}$	Adjacent (± 2 MHz) interference to in-band image frequency @2Mbps		-	TBD	-	dB
$C/I_{\geq 12M,2M,N}$	Adjacent (≥ 12 MHz) interference @2Mbps		-	TBD	-	dB
$C/I_{CO,250K,N}$	Co-Channel interference @250kbps		-	TBD	-	dB
$C/I_{1M,250K,N}$	Adjacent (1 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{2M,250K,N}$	Adjacent (2 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{\geq 3M,250K,N}$	Adjacent (≥ 3 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{Image,250K,N}$	Image frequency interference @250kbps		-	TBD	-	dB
$C/I_{Image \pm 1M,250K,N}$	Adjacent (± 1 MHz) interference to in-band image frequency @250kbps		-	TBD	-	dB
$C/I_{CO,250K,B}$	Co-Channel interference @250kbps		-	TBD	-	dB
$C/I_{1M,250K,B}$	Adjacent (1 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{2M,250K,B}$	Adjacent (2 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{\geq 3M,250K,B}$	Adjacent (≥ 3 MHz) interference @250kbps		-	TBD	-	dB
$C/I_{Image,250K,B}$	Image frequency interference @250kbps		-	TBD	-	dB
$C/I_{Image \pm 1M,250K,B}$	Adjacent (± 1 MHz) interference to in-band image frequency @250kbps		-	TBD	-	dB
$P_{IMD,5TH,1M}$	IMD performance 5 MHz offset @1Mbps		-	TBD	-	dBm
$P_{IMD,5TH,2M}$	IMD performance 10 MHz offset @2Mbps		-	TBD	-	dBm

表 5-11 RSSI 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
$RSSI_{RFC}$	RSSI indication range		-90	-	-15	dBm
$RSSI_{Auu}$	RSSI accuracy		-	± 2	-	dB
$RSSI_{Res}$	RSSI resolution		-	0.25	-	dB
$RSSI_{Per}$	RSSI Sample period		-	0.25	-	us

表 5-12 RF Timing 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
$T_{OSC,EN}$	Crystal oscillator settling time		-	TBD	-	us
$T_{TX,EN}$	Time between TXEN task and READY event after channel FREQUENCY configured		-	TBD	-	us
$T_{RX,EN}$	Time between the RXEN task and READY event after channel FREQUENCY configured in default		-	TBD	-	us

	mode					
$T_{TX,DISABLE}$	Time between DISABLE task and DISABLED event when the radio was in TX		-	TBD	-	us
$T_{RX,DISABLE}$	Time between DISABLE task and DISABLED event when the radio was in RX		-	TBD	-	us
T_{RX-TX}	The time taken to switch from RX to TX or TX to RX		-	150	-	us

表 5-13 RF 功耗特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
$I_{TX,P9dBm,DCDC}$	TX only run current 9dBm @DC-DC		-	13.97	-	mA
$I_{TX,P6dBm,DCDC}$	TX only run current 6dBm @DC-DC		-	9.59	-	mA
$I_{TX,P4dBm,DCDC}$	TX only run current 4dBm @DC-DC		-	6.67	-	mA
$I_{TX,P0dBm,DCDC}$	TX only run current 0dBm @DC-DC		-	5.06	-	mA
$I_{TX,P-4dBm,DCDC}$	TX only run current -4dBm @DC-DC		-	2.56	-	mA
$I_{TX,P-8dBm,DCDC}$	TX only run current -8dBm @DC-DC		-	2.15	-	mA
$I_{TX,P-12dBm,DCDC}$	TX only run current -12dBm @DC-DC		-	1.87	-	mA
$I_{TX,P-16dBm,DCDC}$	TX only run current -16dBm @DC-DC		-	1.64	-	mA
$I_{TX,P-20dBm,DCDC}$	TX only run current -20dBm @DC-DC		-	1.53	-	mA
$I_{TX,P-40dBm,DCDC}$	TX only run current -40dBm @DC-DC		-	1.12	-	mA
$I_{TX,P9dBm,LDO}$	TX only run current 9dBm @LDO		-	35.55	-	mA
$I_{TX,P6dBm,LDO}$	TX only run current 6dBm @LDO		-	16.28	-	mA
$I_{TX,P4dBm,LDO}$	TX only run current 4dBm @LDO		-	12.1	-	mA
$I_{TX,P0dBm,LDO}$	TX only run current 0dBm @LDO		-	9.43	-	mA
$I_{TX,P-4dBm,LDO}$	TX only run current -4dBm @LDO		-	5.74	-	mA
$I_{TX,P-8dBm,LDO}$	TX only run current -8dBm @LDO		-	5.08	-	mA
$I_{TX,P-12dBm,LDO}$	TX only run current -12dBm @LDO		-	4.62	-	mA
$I_{TX,P-16dBm,LDO}$	TX only run current -16dBm @LDO		-	4.04	-	mA
$I_{TX,P-20dBm,LDO}$	TX only run current -20dBm @LDO		-	3.92	-	mA
$I_{TX,P-40dBm,LDO}$	TX only run current -40dBm @LDO		-	3.00	-	mA
$I_{RX,1M,DCDC}$	RX 1Mbps current @DC-DC		-	2.50	-	mA
$I_{RX,2M,DCDC}$	RX 2Mbps current @DC-DC		-	2.71	-	mA
$I_{RX,1M,LDO}$	RX 1Mbps current @LDO		-	4.82	-	mA
$I_{RX,2M,LDO}$	RX 2Mbps current @LDO		-	5.67	-	mA

Test conditions and methods.

1. Transceiver power consumption tested in 1M mode using BLE ADV broadcast mode.
2. 2M mode is used is the power consumption when BLE connection.
3. The power consumption tested is the RF peak power.
4. The test method uses the total power consumption minus the power consumption of the MCU when the RF is not operating to calculate the final power consumption.
5. The sample software tested is based on peripheral_hr.
6. 3.3V Power Supply

5.2.4 GPIO 特性

表 5-14 单个 IO 组合测试

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
V _{IH}	Input high voltage	T _A =25°C	0.7*VDD	-	VDD	V
V _{IL}	Input low voltage	Load capacitance =20pF, T _A =25°C	VSS	-	0.2*VDD	V
V _{HYS}	Input hysteresis voltage, V _{hys} =V _{IH} -V _{IL}	T _A =25°C	-	-	0.3*VDD	V
C _{Iana}	Analog input capacitors	T _A =25°	-	300	-	fF
I _{Lkg}	Leakage current, open-drain mode or input mode	VDD≤VIN≤3.6V	-	-	TBD	uA
R _{PU}	Pull-up resistor	V _{in} =VSS, VDD =3.3V	-	50	-	kΩ
R _{PD}	Pull-down resistor	V _{in} =VSS, VDD =3.3V	-	100	-	kΩ
V _I	Input voltage	T _A =25°C	VSS	-	VDD	V
V _O	Output voltage	T _A =25°C	VSS	-	VDD	V
I _{source}	Source current (Push-pull output) (Except for P03, P15)	V _{in} =VDD-0.5V	8.5	9.5	10.5	mA
	Source current (Push-pull output) (P03, P15)		21	22	23	mA
I _{Sink}	Sink current (Push-pull output)	V _{in} =VSS+0.5V, T _A =25°C	19.5	20.5	21.5	mA
f _{Port_CLK}	IO output frequency	Load capacitance =20pF	-	-	48	MHz

表 5-15 组合测试

Description	Conditions	Status	Remark
IO default state after power on	VDD=3.3V ,T _A =25°C	P00, P01: Pull-up input state Others: High resistance state	
IO status in deepsleep mode	VDD=3.3V ,T _A =25°C	Deepsleep: All gpio retention Standby_m1: All gpio retention Standby_m0: P00, P01, P02 retention	

IO status at reset	VDD=3.3V ,T _A =25°C	P00, P01: Pull-up input state Others: High resistance state	
--------------------	--------------------------------	--	--

表 5-16 nRESET 输入特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
V _{ILR}	Negative threshold voltage, nRESET	VDD=1.8V-3.3V ,T _A =25°C	-	-	0.22*VDD	V
V _{IHR}	Positive threshold voltage, nRESET	VDD=1.8V-3.3V ,T _A =25°C	0.48*VDD	-	-	V
V _{hys_rst}	Schmitt Trigger Voltage Hysteresis	VDD=1.8V-3.3V ,T _A =25°C	-	-	0.26*VDD	V
R _{RST}	nRESET pin internal pull-up resistor	VDD=3.3V ,T _A =25°C	-	7.5	-	kΩ
t _{FR} 0.3pF	nRESET pin input filter pulse time	VDD=3.3V ,T _A =25°C	-	TBD	-	ns

5.2.5 复位特性

表 5-17 复位特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
V _{BOD}	Brown-out detection voltage threshold	BODSEL<2:0> = 000(falling edge), dVDD/dt≤3V/s	-	1.84	-	V
		BODSEL<2:0> = 001(falling edge), dVDD/dt≤3V/s	-	1.96	-	
		BODSEL<2:0> = 010(falling edge), dVDD/dt≤3V/s	-	2.05	-	
		BODSEL<2:0> = 011(falling edge), dVDD/dt≤3V/s	-	2.16	-	
		BODSEL<2:0> = 100(falling edge), dVDD/dt≤3V/s	-	2.25	-	
		BODSEL<2:0> = 101(falling edge), dVDD/dt≤3V/s	-	2.35	-	
		BODSEL<2:0> = 110(falling edge), dVDD/dt≤3V/s	-	2.47	-	
V _{BODhys}	BOD hysteresis voltage	dVDD/dt≤3V/s	100	-	160	mV
T _{BOD_RE1}	BOD response Time(Normal mode)	dVDD/dt≤3V/s	1	32	32	Slow clock
I _{BOD}	BOD operating current	dVDD/dt≤3V/s	-	0.5	-	uA
V _{POR}	Power on reset voltage threshold	Rising edge, dvdd/dt≤3V/s	-	1.71	-	V
		Falling edge, dvdd/dt≤3V/s	-	1.69	-	V
T _{POR}	POR settling time	V _{BAT} =3.3V	-	1.12	-	ms

V_{LVR}	LVR detection voltage threshold	Falling edge, $dVDD/dt \leq 3V/s$	-	1.85	-	V
T_{LVR_RE}	LVR response time	$T_A=25^\circ C$, $dVDD/dt \leq 3V/s$	1	32	32	Slow clock
I_{LVR}	LVR operating current	$T_A=25^\circ C$, $dVDD/dt \leq 3V/s$	-	0.1	-	μA

5.2.6 时钟特性

表 5-18 HXTAL 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f_{HXTL}	High speed crystal oscillator (HXTAL) frequency	$VDD=3.3V$, $T_A=25^\circ C$	-	32	-	MHz
$C_{LoadHXTL}$	Crystal load capacitance	$VDD=3.3V$, $T_A=25^\circ C$	7	9	12	pF
I_{DDHXTL}	HXTAL oscillator operating current	$VDD=3.3V$, $T_A=25^\circ C$	-	250	-	μA
t_{SUHXTL}	HXTAL oscillator startup time	$VDD=3.3V$, $T_A=25^\circ C$, $ESR=40\Omega$, $C_{HXTL}=9pF$	-	465	-	μs
t_{SUHXTL} Quick	HXTAL oscillator Quick startup time	$VDD=3.3V$, $T_A=25^\circ C$, $ESR=40\Omega$, $C_{HXTL}=9pF$	-	155	-	μs
ESR			-	40	-	Ω
$F_{TOLHXTL}$	Frequency tolerance for the crystal	$VDD=3.3V$, $T_A=25^\circ C$	-20	-	20	ppm
PD_{HXTL}	Drive level	$VDD=3.3V$, $T_A=25^\circ C$	-	-	100	μW

表 5-19 LXTAL 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f_{LXTL}	Low speed crystal oscillator (LXTAL) frequency	$VDD=3.3V$, $T_A=25^\circ C$	-	32.768	-	kHz
I_{DDLXTL}	LXTAL oscillator operating current	$VDD=3.3V$, $T_A=25^\circ C$	0.3	0.45	0.67	μA
t_{SULXTL}	LXTAL oscillator Normal startup time	$VDD=3.3V$, $T_A=25^\circ C$	330	550	1500	ms
t_{SULXTL} Quick	LXTAL oscillator Quick startup time	$VDD=3.3V$, $T_A=25^\circ C$	-	125	-	ms
ESR_{LXTL}	Equivalent series resistance $6pF < CL \leq 9pF$	$VDD=3.3V$, $T_A=25^\circ C$	-	70	-	$k\Omega$
$C_{LoadLXTL}$	Crystal load capacitance	$VDD=3.3V$, $T_A=25^\circ C$	-	7	-	pF
PD_{LXTL}	Drive level	$VDD=3.3V$, $T_A=25^\circ C$	-	-	1	μW

表 5-20 32MHz RCH 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f_{IRC32M}	Crystal frequency	VDD=3.3V, $T_A=25^{\circ}\text{C}$	-	32	-	MHz
ACC_{IRC32M}	Frequency accuracy	VDD=3.3V, $T_A=-40^{\circ}\text{C} \sim +105^{\circ}\text{C}$	-	-	-	%
		VDD=3.3V, $T_A=-20^{\circ}\text{C} \sim +85^{\circ}\text{C}$	-	-	-	%
		VDD=3.3V, $T_A=25^{\circ}\text{C}$	-	± 1	-	%
D_{IRC32M}	IRC32M oscillator duty cycle	VDD=3.3V, $f_{IRC32M}=32\text{MHz}$, $T_A=25^{\circ}\text{C}$	48	50	52	%
$I_{DDIRC32M}$	Operating current	VDD=3.3V, $f_{IRC8M}=32\text{MHz}$, $T_A=25^{\circ}\text{C}$	-	82	-	μA
$t_{SUIRC32M}$	Startup time	VDD=3.3V, $f_{IRC32M}=32\text{MHz}$, $T_A=25^{\circ}\text{C}$	-	5	-	μs
df_{IRC32M}	25°C, the frequency drifts with the supply voltage	VDD=1.8V~3.6V, $T_A=25^{\circ}\text{C}$	-	0.5	-	%/V

表 5-21 32kHz RCL 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f_{IRC32k}	Crystal frequency	VDD=3.3V, $T_A=25^{\circ}\text{C}$	-	32	-	kHz
ACC_{IRC32K}	Frequency accuracy	VDD=3.3V, $T_A=25^{\circ}\text{C}$ (After calibration)	-	± 500	-	ppm
D_{IRC32K}	IRC32K oscillator duty cycle	VDD=3.3V, $f_{IRC32K}=32\text{kHz}$, $T_A=25^{\circ}\text{C}$	48	50	52	%
$I_{DDIRC32K}$	Operating current	VDD=3.3V, $f_{IRC8K}=32\text{kHz}$, $T_A=25^{\circ}\text{C}$	-	310	-	nA
$t_{SUIRC32K}$	Startup time	VDD=3.3V, $f_{IRC32K}=32\text{kHz}$, $T_A=25^{\circ}\text{C}$	-	480	-	μs
df_{IRC32K}	25°C, The frequency drifts with the supply voltage	VDD=1.8V~3.6V, $T_A=25^{\circ}\text{C}$	-	1	-	%/V

表 5-22 DPLL 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
f_{PLLIN}	PLL input clock frequency	VDD=3.3V, $T_A=25^{\circ}\text{C}$	-	32	-	MHz
f_{PLL}	PLL output clock frequency	VDD=3.3V, $T_A=25^{\circ}\text{C}$	48	48	48	MHz
I_{PLL}	Operating current	VDD=3.3V, $T_A=25^{\circ}\text{C}$	96	105	145	μA

5.2.7 ADC 特性

表 5-23 电源及输入范围条件

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
$V_{Ax(VBG\ adc)}$	Analog input voltage range, VBG (1.2V)	VDD=3.3V, $T_A=25^{\circ}C$	0	-	1.2	V
$V_{Ax(VDD)}$	Analog input voltage range, VDD	VDD=3.3V, $T_A=25^{\circ}C$	0	-	VDD	V
I_{ADC}	ADC supply current	VDD=3.3V, $T_A=25^{\circ}C$ Fadc=16MHz	-	0.55	-	mA
C_{sample}	Internal sample and hold capacitors (PAD and PCB capacitors not included)		-	10	-	pF
R_{ADC}	Sampling switch resistance	$0V \leq V_{Ax} \leq VDD$	-	300	-	Ω
R_{In}	External input impedance, continuous sampling	$0V \leq V_{Ax} \leq VDD$	0.86	-	4734.70	k Ω

表 5-24 ADC 内置电压基准

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
VBG_{ADC}	Internal 1.2V Reference Voltage	VDD=3.3V, $T_A=25^{\circ}C$	1.1	1.2	1.3	V
T_{Coef}	Temperature factor	$T_A=-40^{\circ}C \sim 105^{\circ}C$; VDD=1.8V~3.6V	-	30	-	ppm/ $^{\circ}C$

表 5-25 时间参数

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
F_{ADC}	ADC clock frequency	VDD=3.3V, $T_A=25^{\circ}C$	4	16	24	MHz
T_S	Sample time	VDD=3.3V, $T_A=25^{\circ}C$	4	1539	8192	1/F _{adc}
T_{CONV}	Conversion time	VDD=3.3V, $T_A=25^{\circ}C$	32	1580	8298	1/F _{adc}

表 5-26 线性度参数

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
INL	Integral linearity error	VDD=3.3V, $T_A=25^{\circ}C$	-	-	± 3	LSB
DNL	Differential linearity error	VDD=3.3V, $T_A=25^{\circ}C$	-	-	± 2	LSB

SNR	Signal to Noise Ratio	Fadc = 16MHz Input Clock 250kHz VDD=3.3V, T _A =25°C	-	64.3	-	dB
THD	Total harmonic distortion		-	75	-	dB
SFDR	Spurious-free signal dynamic range		-	77.29	-	dB
ENOB	Effective number of bits		-	10.33	-	Bit

表 5-27 RIN

ADC significant bit	FADC(MHz)	Ts(cycles)	Ts(us)	Rinmax(kΩ)
12	32	4	0.125	0.86
12	32	8	0.25	2.01
12	32	32	1	8.95
12	32	64	2	18.20
12	32	128	4	36.69
12	32	8192	256	2367.20
12	16	4	0.25	2.01
12	16	8	0.5	4.32
12	16	32	2	18.20
12	16	64	4	36.69
12	16	128	8	73.68
12	16	8192	512	4734.70

Note: The sampling condition is continuous sampling.

5.2.8 PMU 特性

表 5-28 PMU 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
V _{HLDO}	HLDO output voltage range, external capacitor	VDD=3.3V, T _A =25°C	1.1	1.2	1.4	V
VDD _{PSRR}	Power supply rejection ratio of VDD	VDD=3.3V, T _A =25°C	-15	-	-	dB

5.2.9 DCDC 特性

表 5-29 DCDC 特性

Symbol	Description	Conditions	Parameter			Unit
			Min	Typ	Max	
V _{IN_DCDC}	Input voltage range	VDD=3.3V, T _A =25°C	2	-	3.6	V
V _{OUT_DCDC}	Output voltage range	VDD=3.3V, T _A =25°C	1.2	1.5	2	V
T _{EN_DCDC}	Standup time	VDD=3.3V, I _{LOAD} =10mA, T _A =25°C, L _{DCR} =80mΩ	-	200	-	us

η	Efficiency	VDD=3.3V, $I_{LOAD}=10mA$, $T_A=25^\circ C$, $L_{DCR}=80m\Omega$	-	88	-	%
VRPL _{DCDC}	Ripple	VDD=3.3V, $T_A=25^\circ C$	-	43	-	mv
I_{OUT}	Drive peak current	VDD=3.3V, $T_A=25^\circ C$	-	100	-	mA
I_{AVG}	Drive average current	VDD=3.3V, $T_A=25^\circ C$	-	30	-	mA
L_{DCDC}	Effective inductance	VDD=3.3V, $T_A=25^\circ C$	-	2.2	-	μH
C_{OUT_DCDC}	Effective load capacitance	VDD=3.3V, $T_A=25^\circ C$	1	4.7	-	μF
F_{osc_DCDC}	Oscillation frequency	VDD=3.3V, $T_A=25^\circ C$	-	-	1000	kHz

5.2.10 MCU 电流特性

Symbol	Parameter	Conditions		DCDC OFF	DCDC ON
				Typ(mA)	Typ(mA)
Run mode	All peripherals clock on, run while(1) in flash	System clock source: RCH (.cal 32M)	4M	0.91	0.686
			8M	1.3	0.916
			16M	2.09	1.36
			32M	3.77	2.28
		System clock source: XTH (.off rch)	4M	1.42	0.985
			8M	1.81	1.2
			16M	2.61	1.64
			32M	4.29	2.54
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 48M)	6M	1.56	1.05
			12M	2.15	1.38
			24M	3.4	2.05
			48M	5.71	3.44
		System clock source: DPLL (.ref xth) (.base 48M) (.off rch)	6M	2.08	1.34
			12M	2.69	1.67
			24M	3.95	2.34
			48M	6.18	3.57
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 64M)	4M	1.39	0.953
			8M	1.78	1.17
			16M	2.58	1.61
			32M	4.27	2.51
		System clock source: DPLL (.ref xth) (.base 64M) (.off rch)	4M	1.9	1.25
			8M	2.3	1.46
			16M	3.13	1.91
			32M	4.85	2.82
	All peripherals clock off, run while(1) in flash	System clock source: RCH (.cal 32M)	4M	0.736	0.523
			8M	0.943	0.719
			16M	1.37	0.963
			32M	2.23	1.43
		System clock source: XTH (.off rch)	4M	0.945	0.723
			8M	1.15	0.84
			16M	1.57	1.07

			32M	2.42	1.54
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 48M)	6M	1.01	0.747
			12M	1.32	0.921
			24M	1.96	1.25
			48M	3.29	2.00
		System clock source: DPLL (.ref xth) (.base 48M) (.off rch)	6M	1.22	0.903
			12M	1.54	1.05
			24M	2.19	1.41
			48M	3.54	2.12
		System clock source: DPLL (.ref rch(.cal 32M)) (.base 64M)	4M	0.925	0.693
			8M	1.14	0.818
			16M	1.56	1.05
			32M	2.43	1.53
		System clock source: DPLL (.ref xth) (.base 64M) (.off rch)	4M	1.14	0.831
			8M	1.35	0.947
			16M	1.78	1.18
			32M	2.66	1.66

Notes: Test conditions - DVDD=1.2V, VDD=3.3V, VBG=1.2V, T_A=25°C

Symbol	Parameter	Conditions		Typ(μA)
standby m0	wake by gpio			0.28
Symbol	Parameter	Conditions	Clk source	Typ(μA)
standby m1	wake by 32k timer	sw=0x0d (Note 1)	XTL	1.88(Note 3)
		sw=0x1f (Note 1)	XTL	2.18(Note 3)
	wake by gpio	sw=0x0d (Note 1)		1.32(Note 3)
		sw=0x1f (Note 1)		1.63(Note 3)
	wake by wdt	sw=0x0d (Note 1)	XTL	1.85(Note 3)
		sw=0x1f (Note 1)	XTL	2.18(Note 3)
	wake by lvr	sw=0x0d (Note 1)		2.05(Note 3)
		sw=0x1f (Note 1)		2.42(Note 3)
	wake by bod	sw=0x0d (Note 1)		2.11(Note 3)
		sw=0x1f (Note 1)		2.48(Note 3)
Deepsleep	wake by 32k timer	sw=0x0d (Note 1)	XTL	2.96(Note 3)
		sw=0x1f (Note 1)	XTL	3.37(Note 3)
	wake by gpio	sw=0x0d (Note 1)		2.46(Note 3)
		sw=0x1f (Note 1)		2.86(Note 3)
	wake by wdt	sw=0x0d (Note 1)	XTL	2.99(Note 3)
		sw=0x1f (Note 1)	XTL	3.37(Note 3)
	wake by peripheral timer	sw=0x0d (Note 1)	XTL	4.03(Note 3&4)
		sw=0x1f (Note 1)	XTL	4.73(Note 3)
	wake by 32k timer	sw=0x0d (Note 1)	XTL*	4.06(Note 3&4)
		sw=0x1f (Note 1)	XTL*	4.69(Note 3)

Sleep	wake by gpio	LDO all peripheral clocks on	4620
		LDO non-essential peripheral clocks off	1820
		DCDC all peripheral clocks on	2700
		DCDC non-essential peripheral clocks off	1210

Note: Test conditions - DVDD=1.2V, VDD=3.3V, VBG=1.2V, TA=25°C, DCDC ON

Note*: PWM output enable.

Notes:

1. The **sw** indicates the module control switch for PAN107x power retention in low-power mode, bit 1 for power retention, bit 0 for power down, 0d = sram32k+phy_sram+cpu_retention, 1f = all_sram_retention
2. In LPLDOH Enhance mode, the lpddoh voltage is affected by lpddoh trim and vref trim, where vref trim compensates lddoh when lddoh jitter, so that it does not fall to the voltage of vref trim as much as possible.
3. In LPLDOH mode2, the lpddoh trim voltage does not take effect, and the effective voltage is the vref_trim voltage. The lpddoh undershoot does not occur in this mode, and the bottom current is reduced.
4. PWM can output waveforms normally in *Deepsleep* mode, but only in *Deepsleep* mode2, and see Note 5 for instructions on *Deepsleep* mode2.
5. When the register *LP_FL_CTRL* [31] is 1, the switch between LPLDOL and LPLDOH is connected, and LPLDOL enable is off. LPLDOH supplies power to LPLDOL. This is the mode 2. Mode 2 power consumption increases.

6 参考原理图

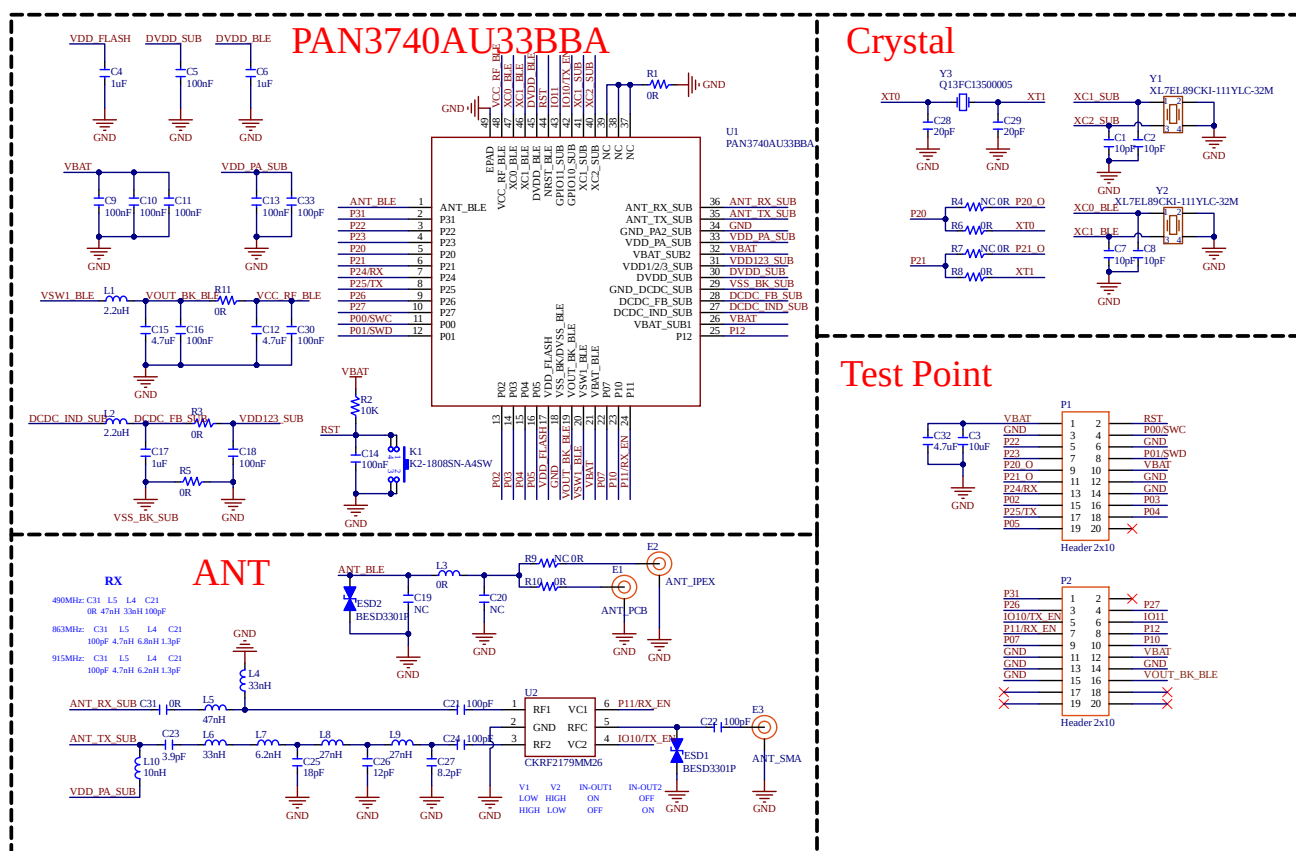


图 6-1 参考原理图

7 封装信息

7.1 QFN48 封装

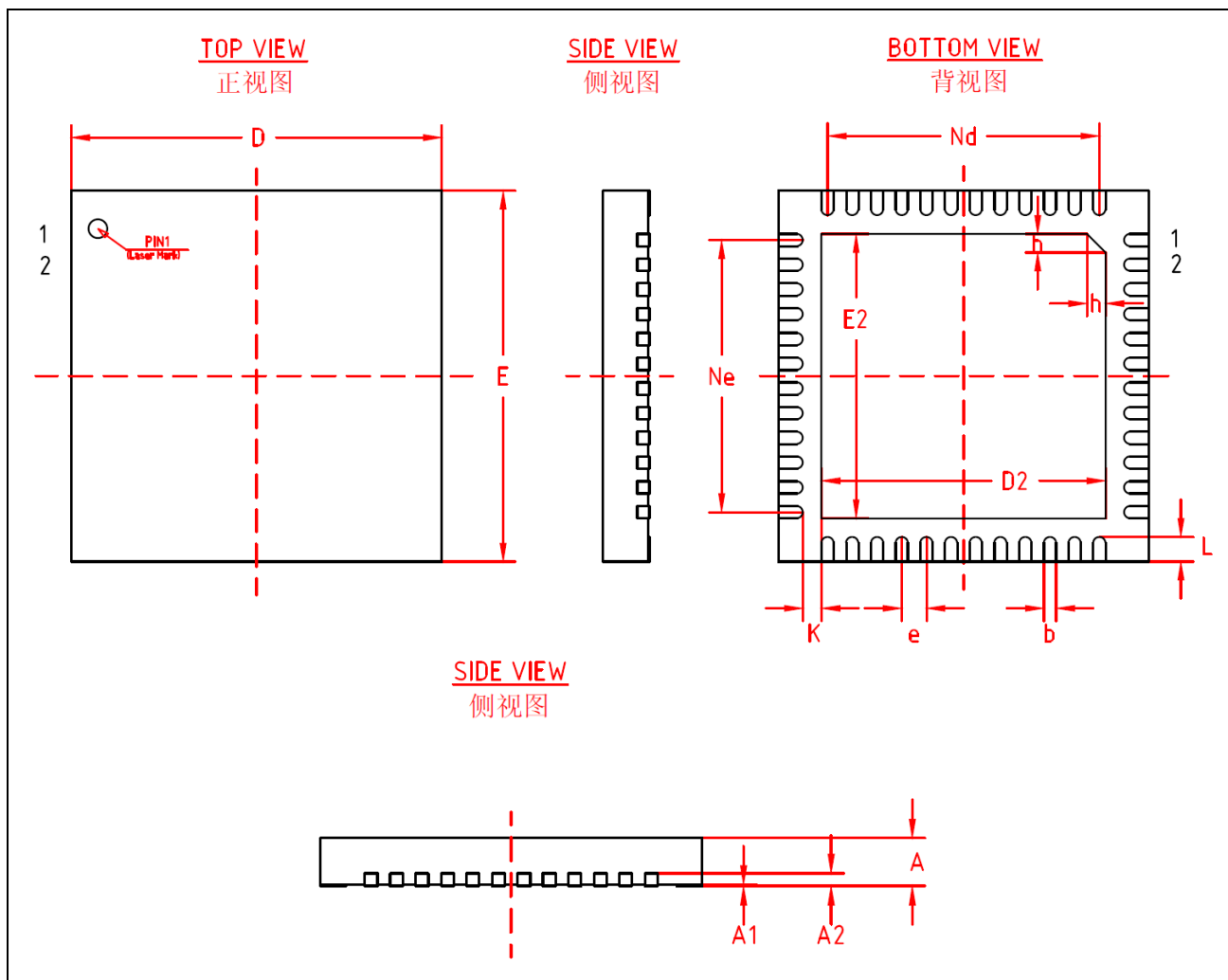


图 7-1 封装图

表 7-1 封装尺寸

SYMBOL	MIN (mm)	NOM (mm)	MAX (mm)
A	0.70	0.75	0.80
A1	-	0.02	0.05
A2	0.203 REF		
b	0.15	0.20	0.25
D	5.90	6.00	6.10
D2	4.55	4.60	4.65
E	5.90	6.00	6.10
E2	4.55	4.60	4.65
e	0.40 BSC		
K	0.25	0.30	0.35

L	0.35	0.40	0.45
h	0.25	0.30	0.35
Ne	4.40 BSC		
Nd	4.40 BSC		

缩略语

ACK	应答信号	MCU	微控制器单元
ADC	模数转换器	MISO	主入从出
bandgap	带隙基准	MM	带电机器的放电模式
BLE	低功耗蓝牙	MOSI	主出从入
BOD	掉电检测	PLL	锁相环
BQB	蓝牙认证	POR	上电复位
Cache	高速缓冲存储器	PWM	脉冲宽度调制
CAD	信道活跃检测	QDEC	正交解码器
CDM	带电器件模型	RAM	随机存取存储器
Chirp	线性调频	trg	触发
CS	片选	OSC	振荡器
CTS	接收请求	PA	功率放大器
DAC	数模转换器	RC	电阻电容晶振
DCDC	降压式直流变换器	RF	射频
DMA	直接内存访问	PLL	锁相环
DPLL	数字锁相环	RSSI	接收信号强度
eFuse	一次性可编程存储器	RX	接收
ECC	椭圆加密算法	SDA	数据信号线
ESD	静电释放	SF	扩频因子
ext	外部 IO 口	SPI	串行外设接口
ETSI	欧洲电信标准化协会	SRAM	静态随机存取存储器
FCC	美国联邦通信委员会	SWD	串行线调试
FMC	flash 控制器	STB	待机模式
FIFO	先入先出	TEMP	温度传感器
GPIO	通用 I/O	TX	发射
HBM	带电的人体的放电模式	RTS	发送请求
HLDO	强驱动低压差线性稳压器	UART	通用异步接收器/发送器
I2C	两线式串行总线	WDT	看门狗定时器
IAP	应用编程	WWDT	串口看门狗定时器
IRQ	中断请求	XTAL	外部晶振
Latch-up	闩锁效应	VCO	压控振荡器
LDO	低压差稳压器		
LED	发光二极管		
LVR	低电压复位		
MAC	介质访问控制层		

修订历史

版本	日期	内容
1.0	Feb.2024	初版发布
1.1	Jun.2024	更新为 QFN48
1.2	Jan.2025	更新参考原理图，更新工作频段和功耗

此版本为内部版本，仅供参考。

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